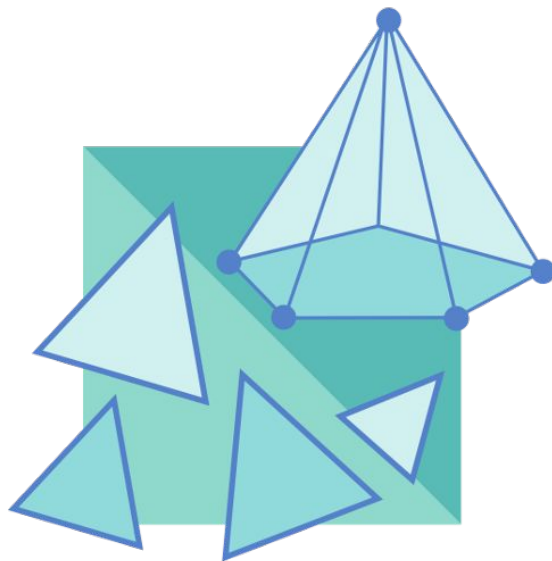


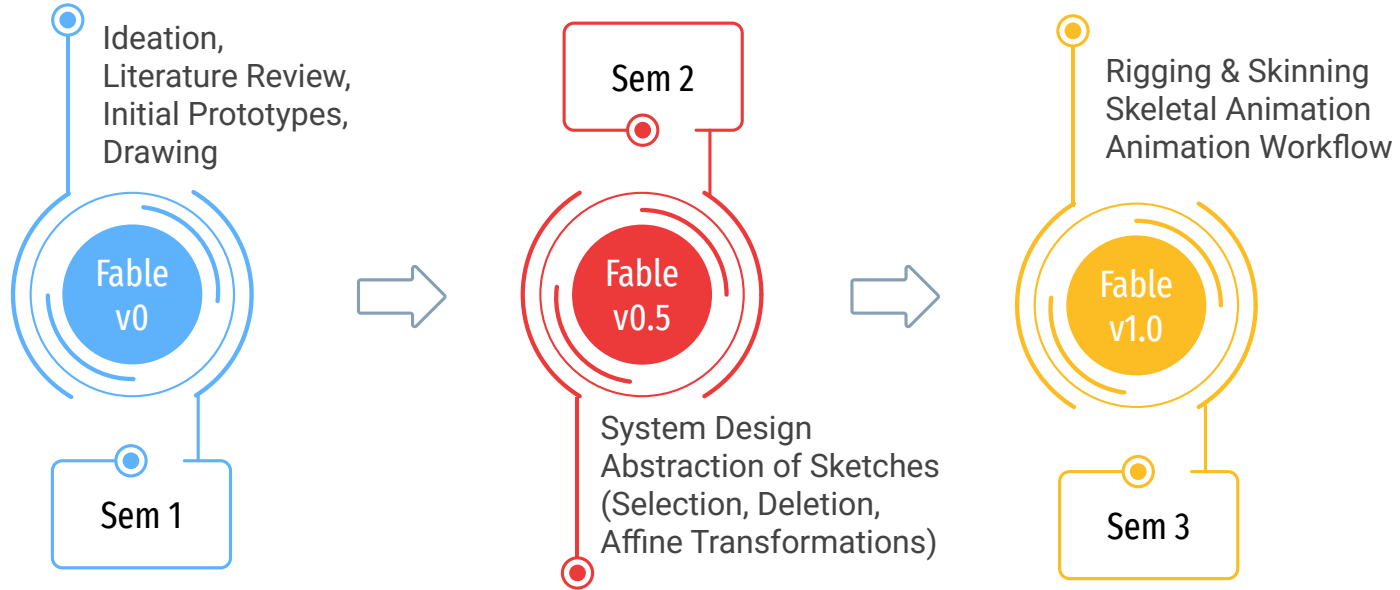


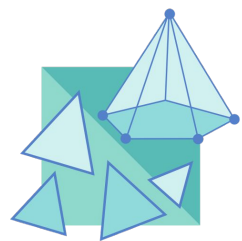
Graphics Research Group

Sketch Based Animation in 2D

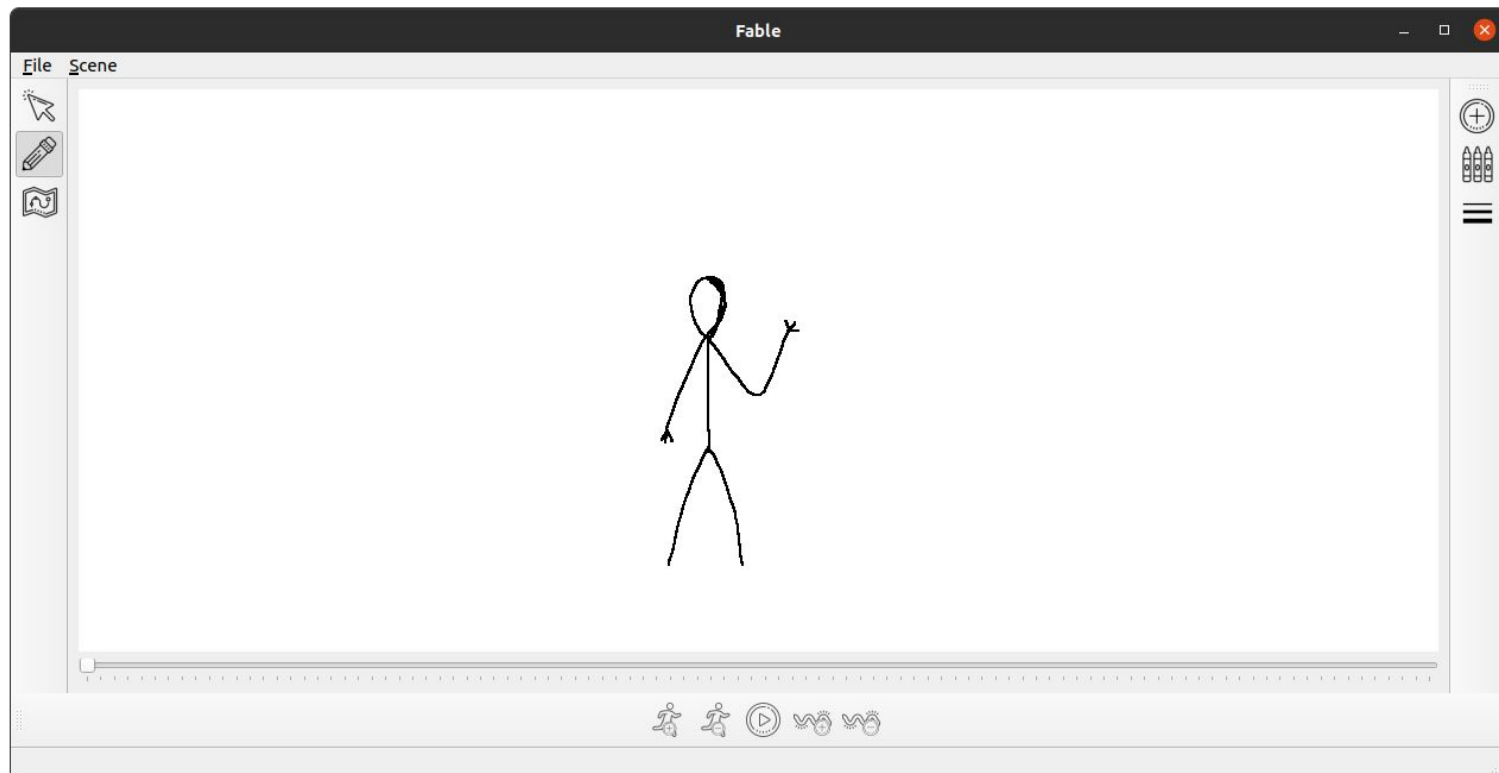
Aditya Singh Rathore
Shrikant Garg

Timeline & Accomplishments

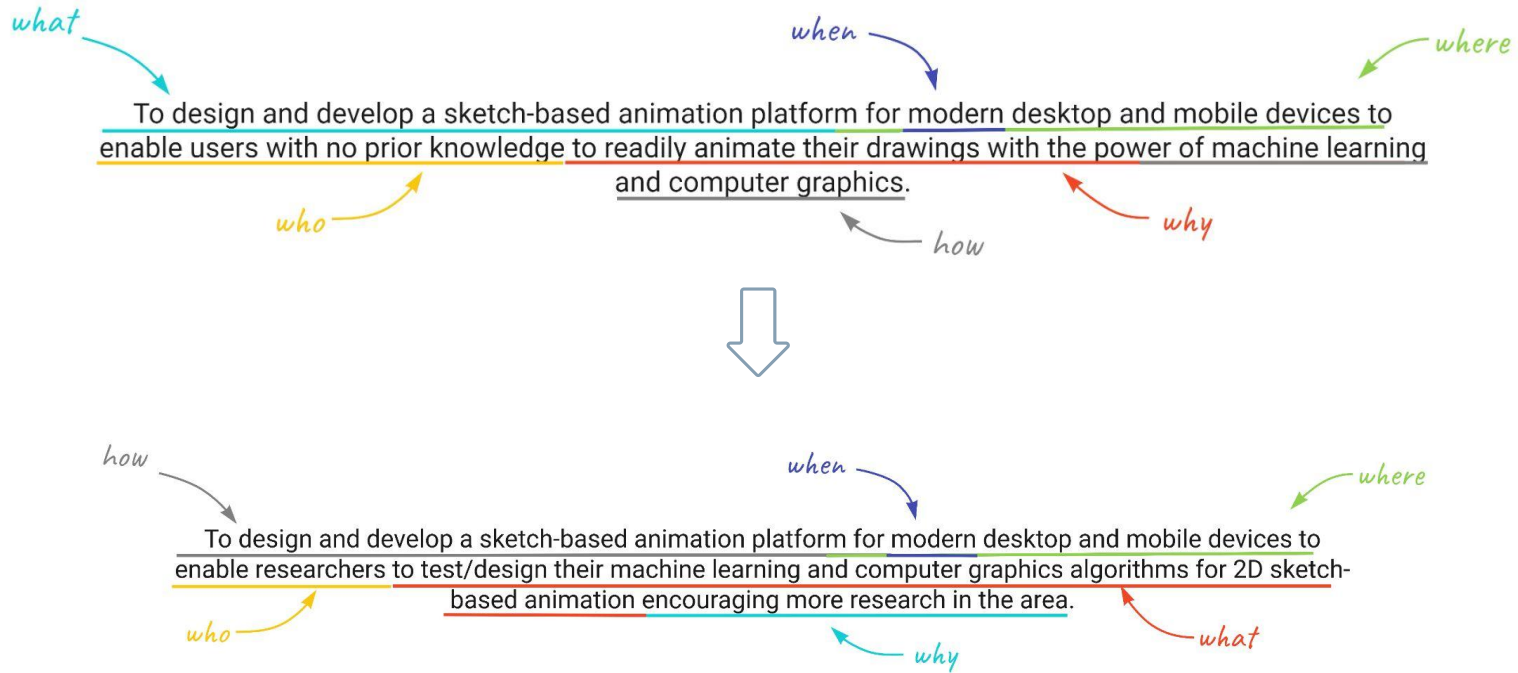


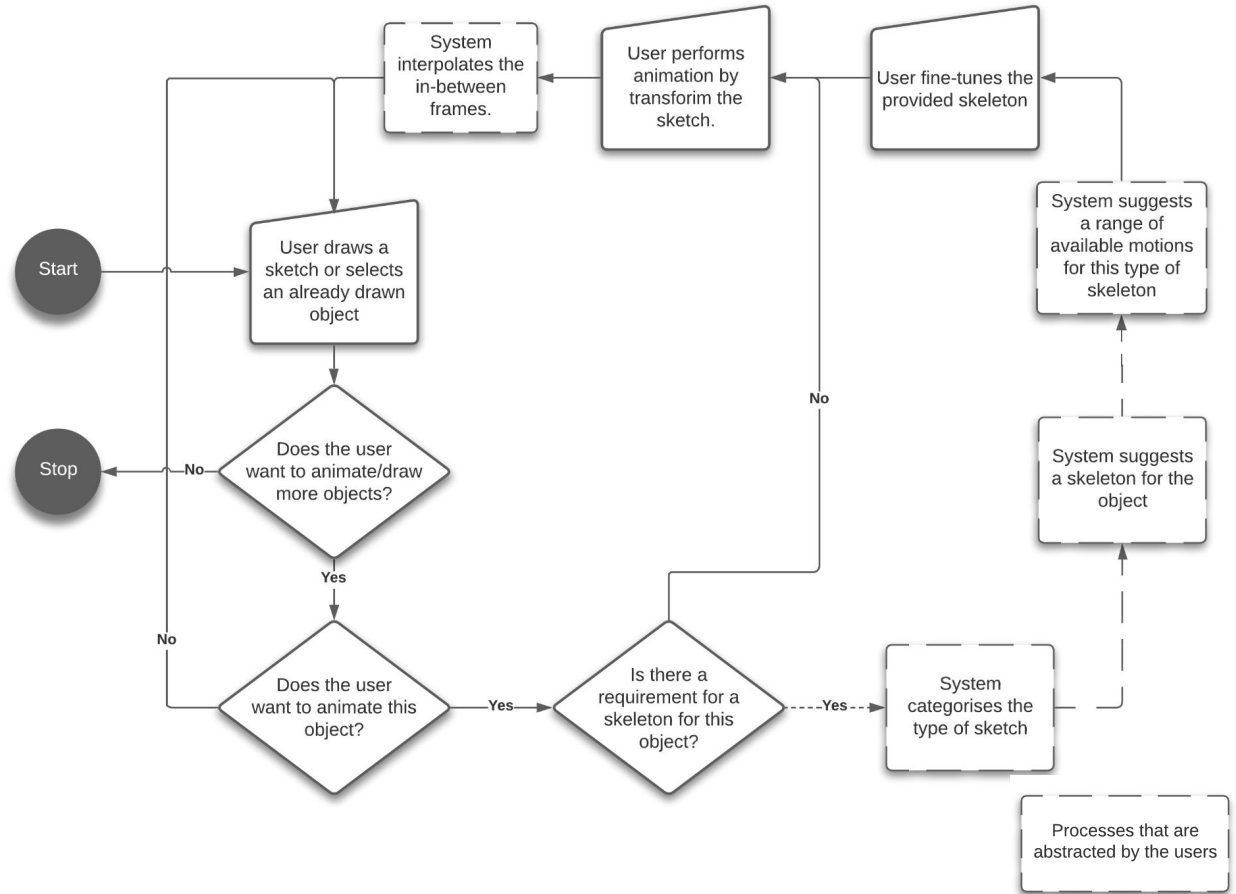
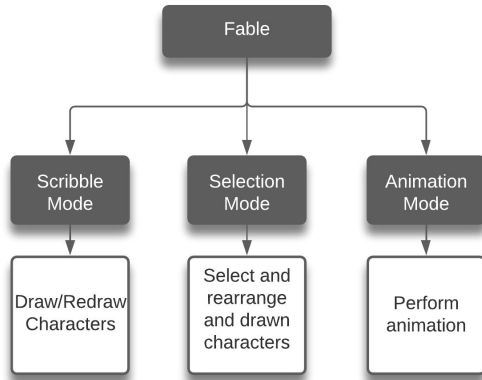


Fable



Problem Statement

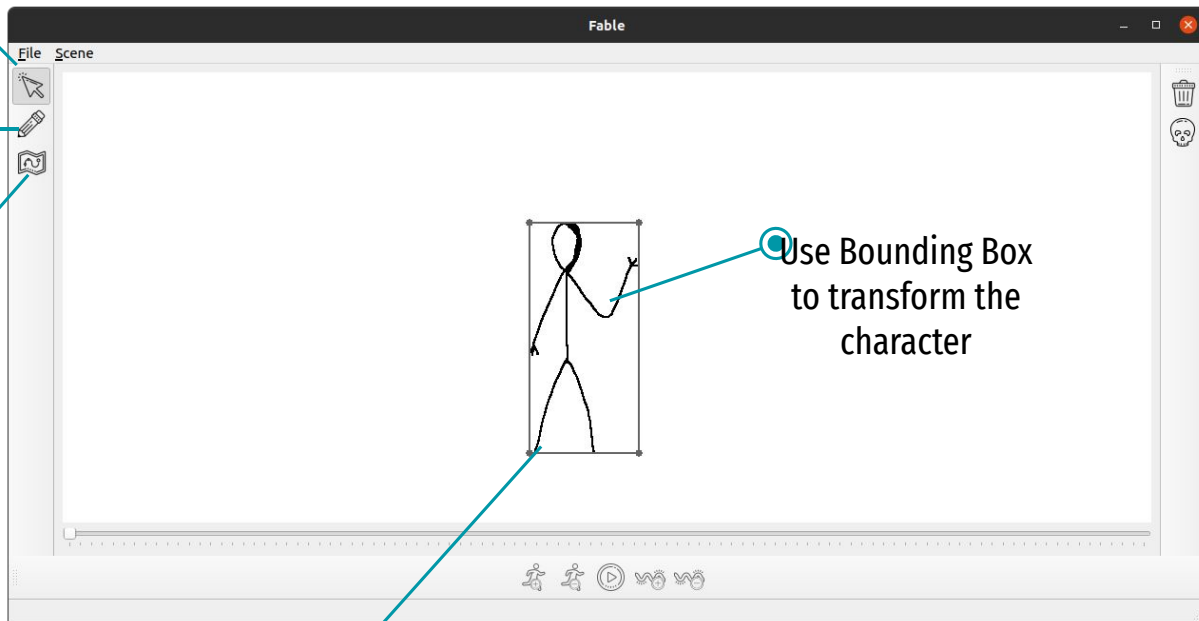




Selection
Mode

Scribble
Mode

Animation
Mode

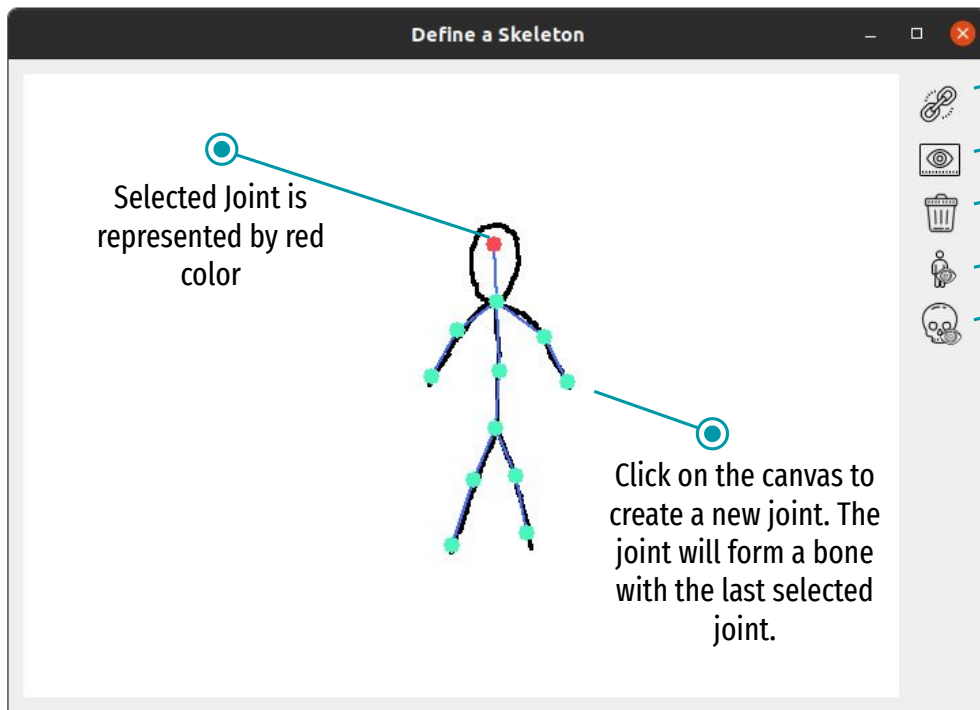


Delete
Selected
Character

Define
Skeleton
Selected
Character

Use Bounding Box
to transform the
character

Left click on Character to select



Selected Joint is represented by red color

Click on the canvas to create a new joint. The joint will form a bone with the last selected joint.



Bind the skeleton to character



Preview Rigging and Skinning



Delete Selected Joint

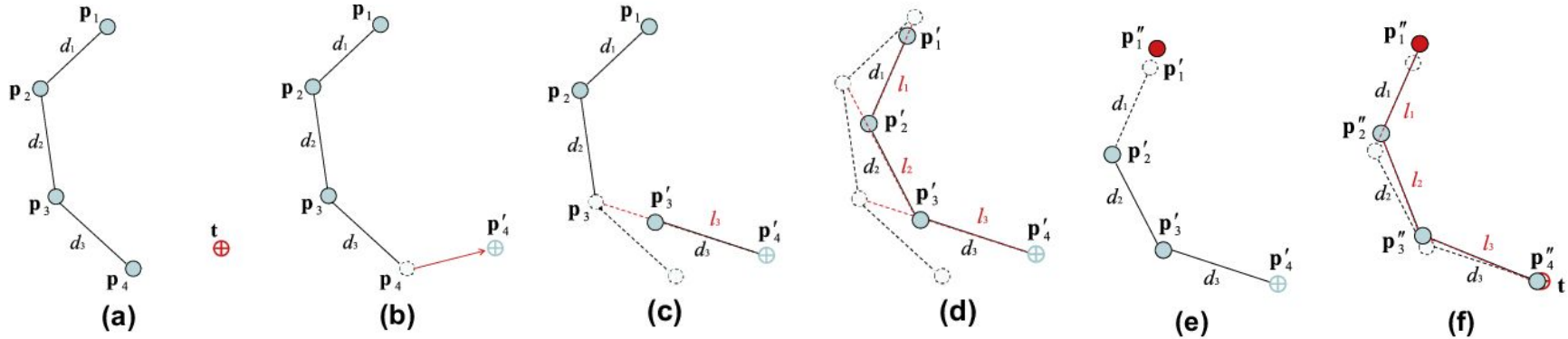


Hide/Show Skeleton



Hide/Show Cage

Inverse Kinematics (FABRIK)



Forward and Backward Inverse Reaching Kinematics

Inverse kinematics is the mathematical process of calculating the variable joint parameters needed to place the end of a kinematic chain, such as a robot manipulator or animation character's skeleton, in a given position and orientation relative to the start of the chain. Given joint parameters, the position and orientation of the chain's end, e.g. the hand of the character or robot, can typically be calculated directly using multiple applications of trigonometric formulas, a process known as forward kinematics. However, the reverse operation is, in general, much more challenging.

Live Demonstration (FABRIK)

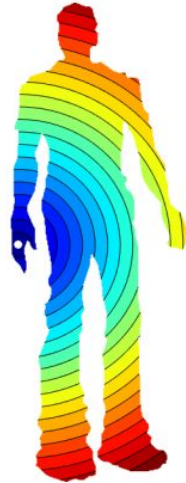
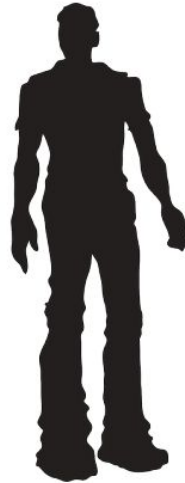
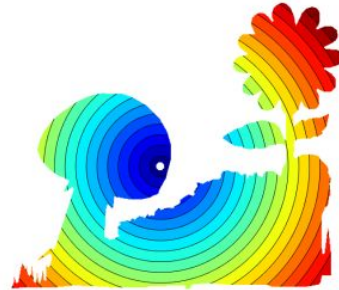
Linear blend Skinning

- Deform the sketch using skeletal structure.
- Each point of sketch is assigned to multiple bones.
- Use the transform of the associated bone to position each vertex of the skin in world space T_i
- Each bone i to which vertex v_j belongs to is assigned a nonzero weight w_{ij}
- The world space position of the vertex is computed as the weighted average of the world space positions obtained from each bone via rigid skinning:

$$v'_j = \sum_i w_{ij} T_i v_j^i$$

Computing Weights

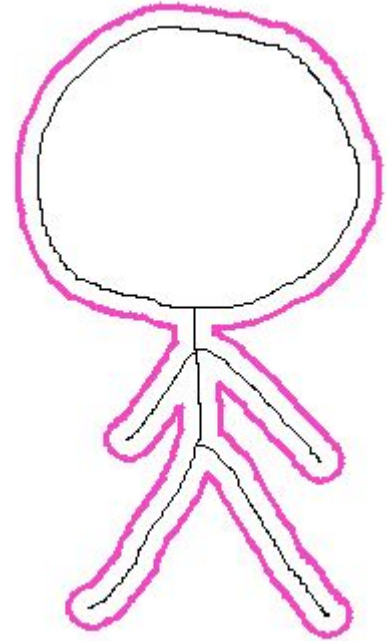
- Used an algorithm to calculate weights for each vertex and all its associated bones based on internal distances.



Ojaswa Sharma and Ranjith Tharayil.
2015. Shape-aware MLS deformation for
line handles. In SIGGRAPH Asia 2015
Technical Briefs (SA '15). Association
for Computing Machinery, New York, NY,
USA, Article 20, 1-4.
DOI:<https://doi.org/10.1145/2820903.2820908>

Boundary of a Sketch ?

- We have used Offset curves to generate boundary of a sketch.
- Offset curve is locus of all points which are at a fixed distance d from the the curve.
- We have used [clipperlib](#)

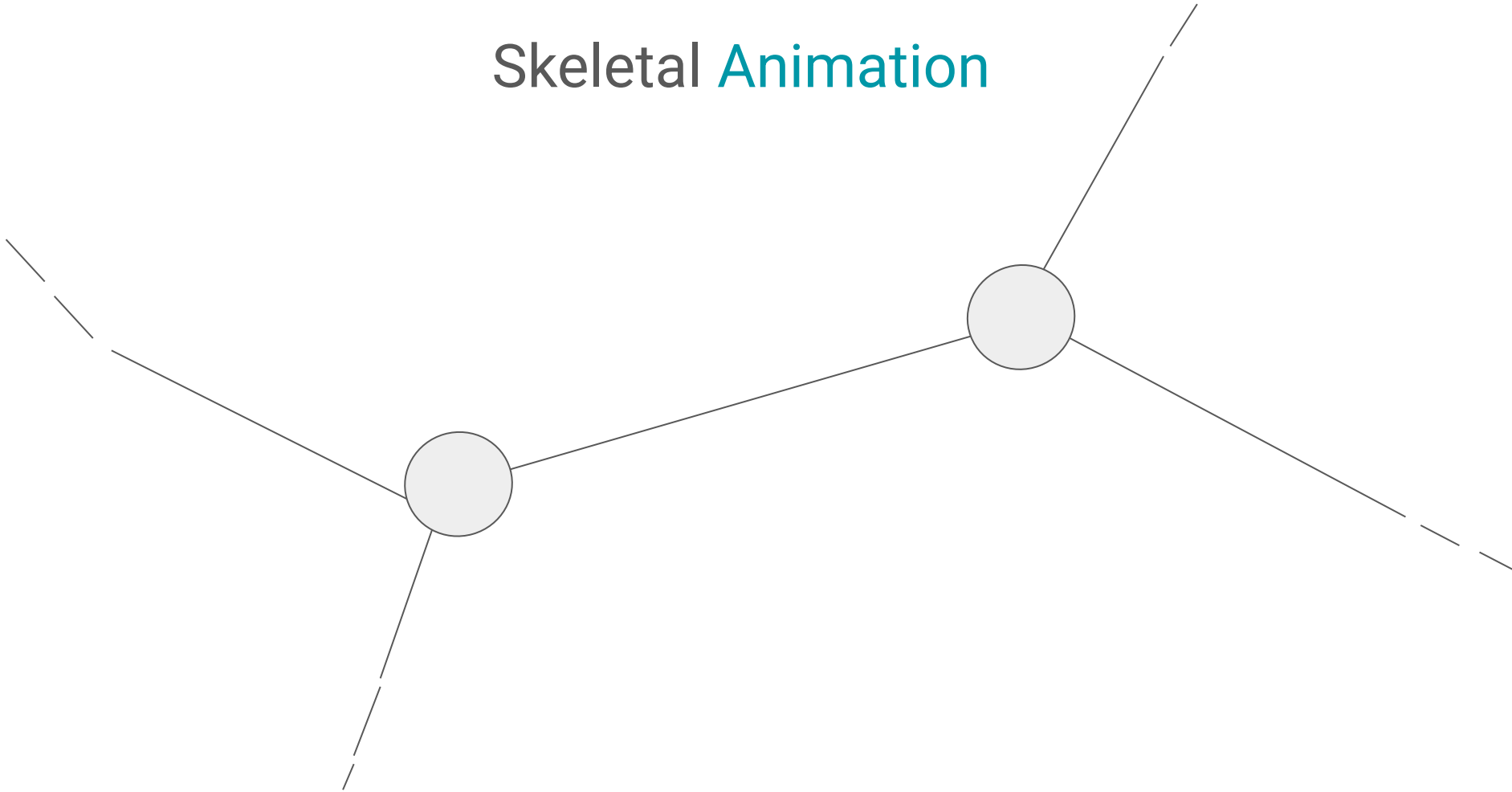


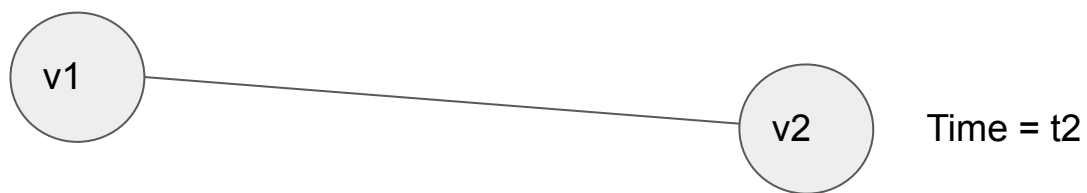
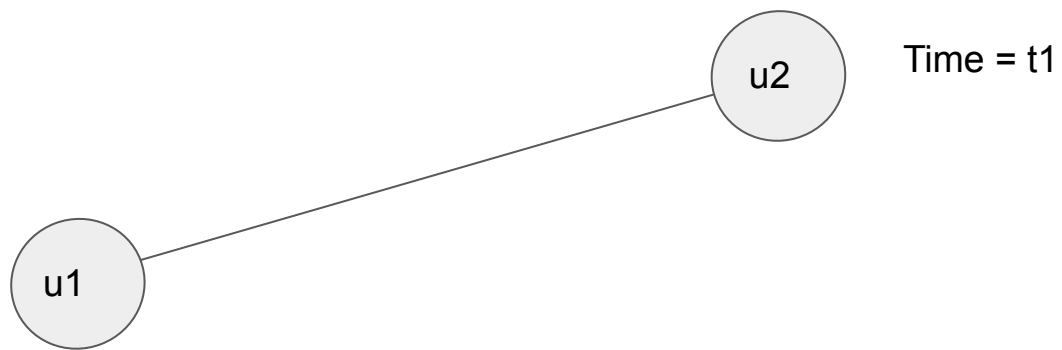
Live Demonstration (Skinning)

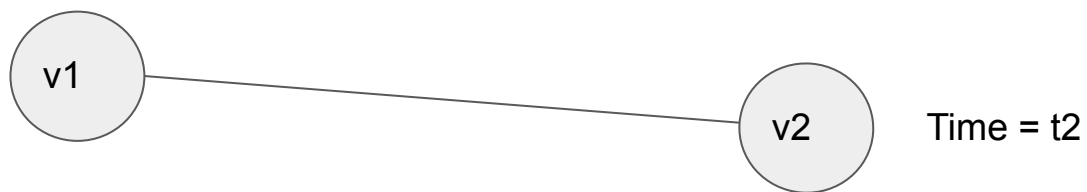
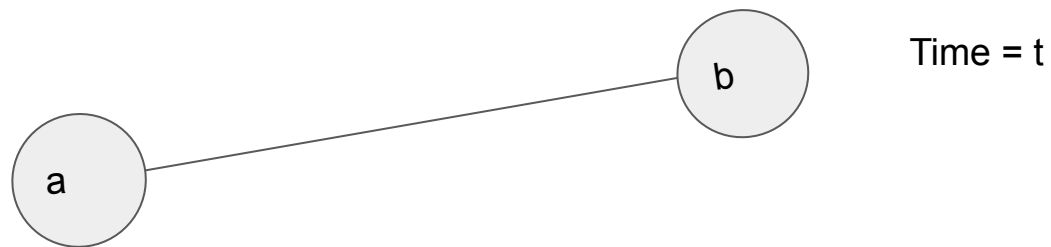
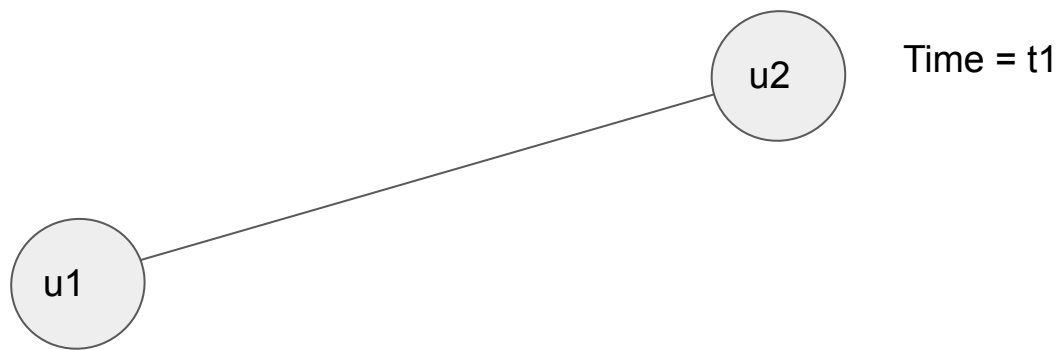
Skeletal Animation

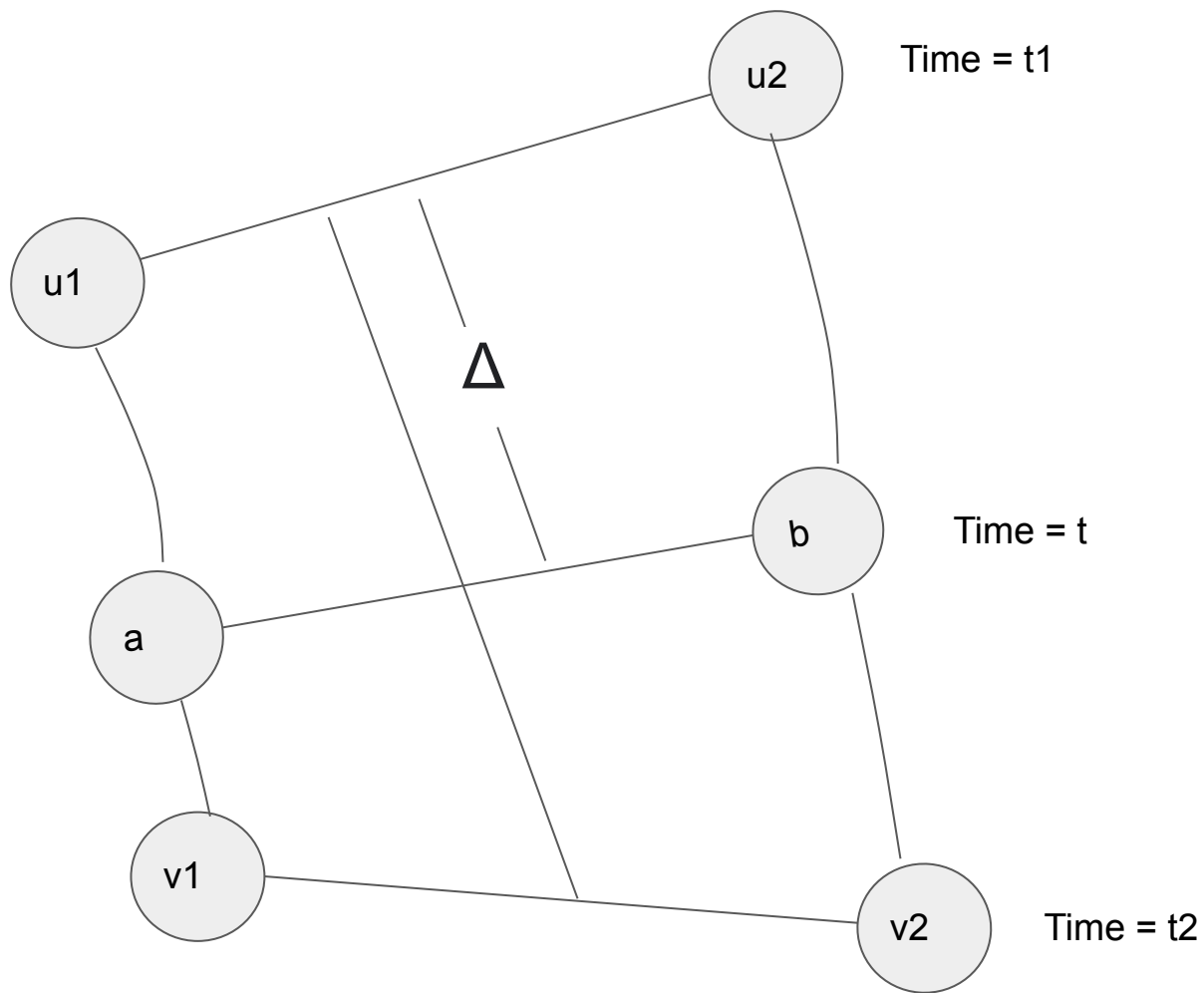
- Given poses of a skeleton at times $t_1, t_2, t_3, \dots t_n$.
- Given a time t , such that $t \geq t_1$ and $t \leq t_n$, calculate the skeleton position using linear interpolation.
- Use skinning to transform points of sketch to animate the character.

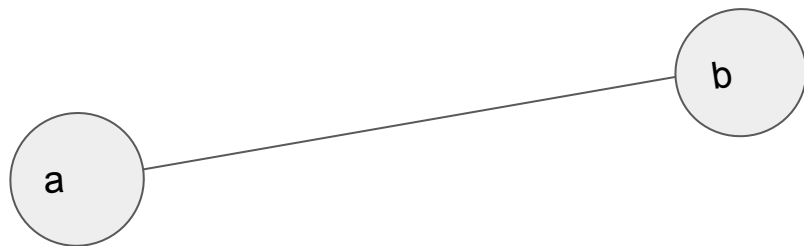
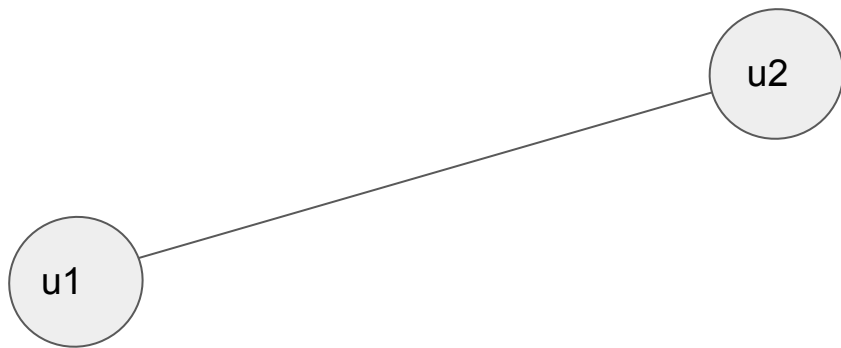
Skeletal Animation

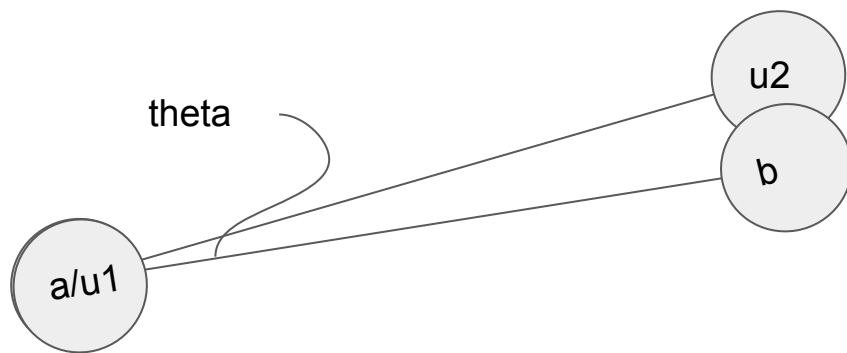


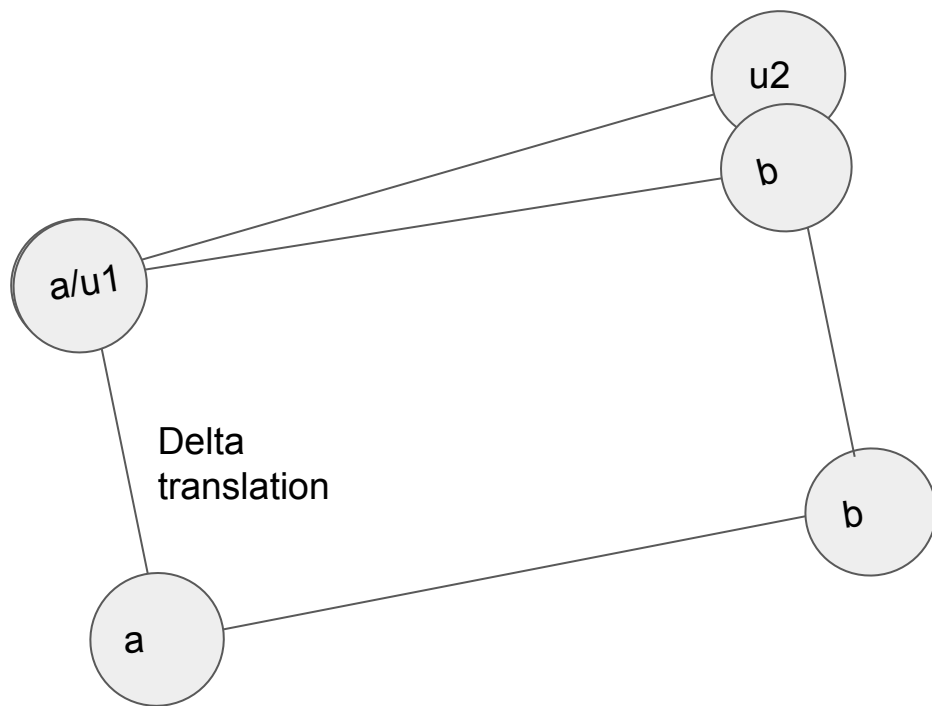






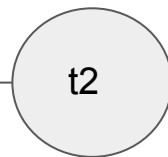
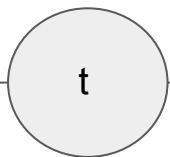
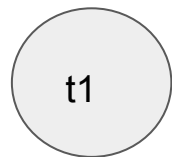






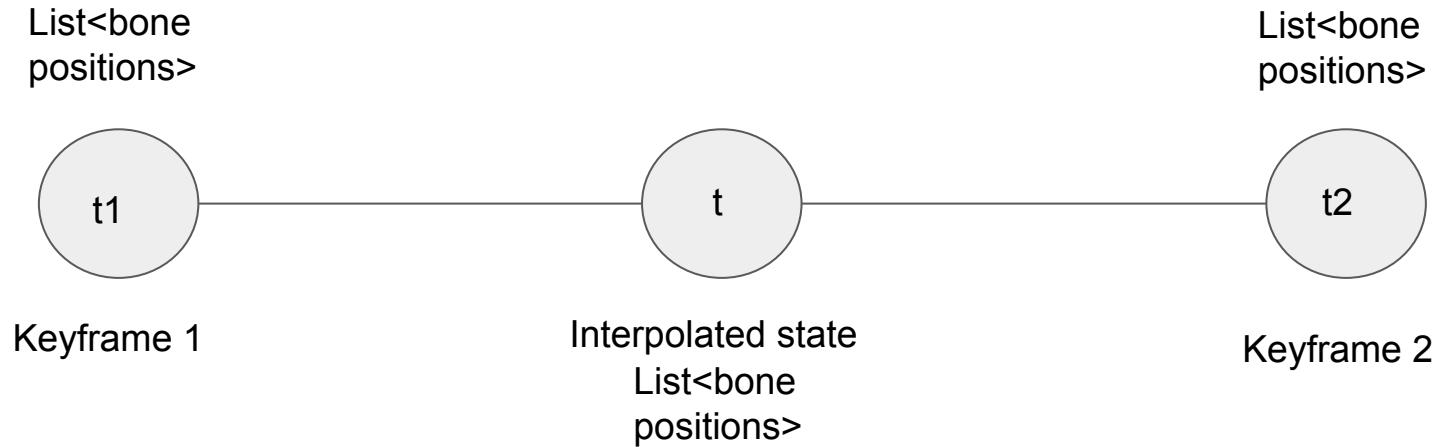
List<bone
positions>

List<bone
positions>



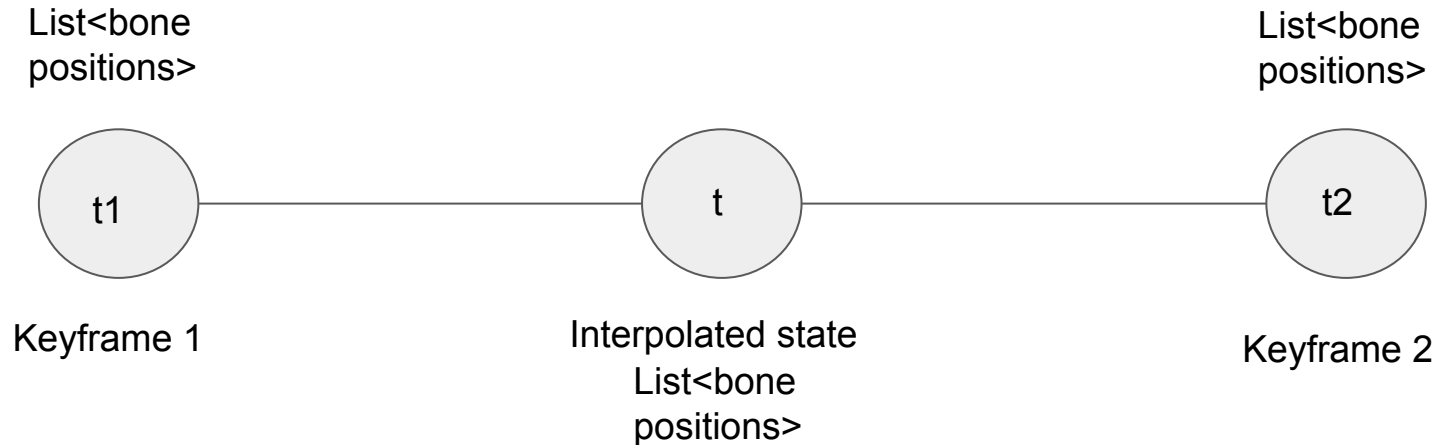
Keyframe 1

Keyframe 2

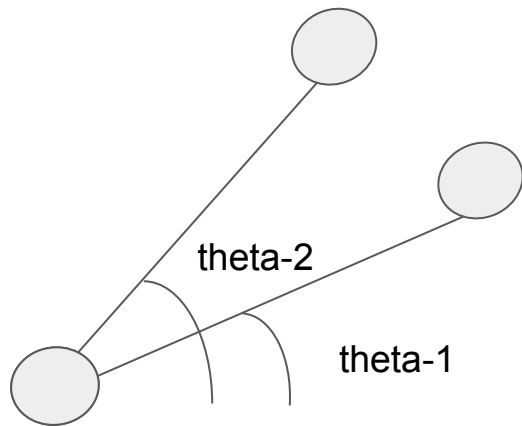


$$\textit{lerp}(a, b, t) = \underset{\text{Start}}{a} + \left(\underset{\text{Distance from A to B}}{b - a} \right) * \underset{\text{Percentage}}{\Delta}$$

$$\Delta = \frac{t - t_1}{t_2 - t_1}$$



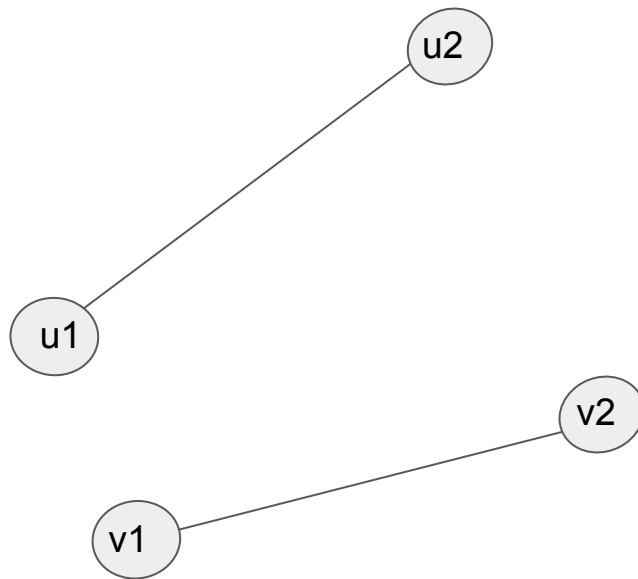
Theta



$$\text{Theta} = \text{lerp}(\text{theta-1}, \text{theta-2}, \Delta)$$

$$\text{theta-i} = \text{arctangent}(\Delta y, \Delta x)$$

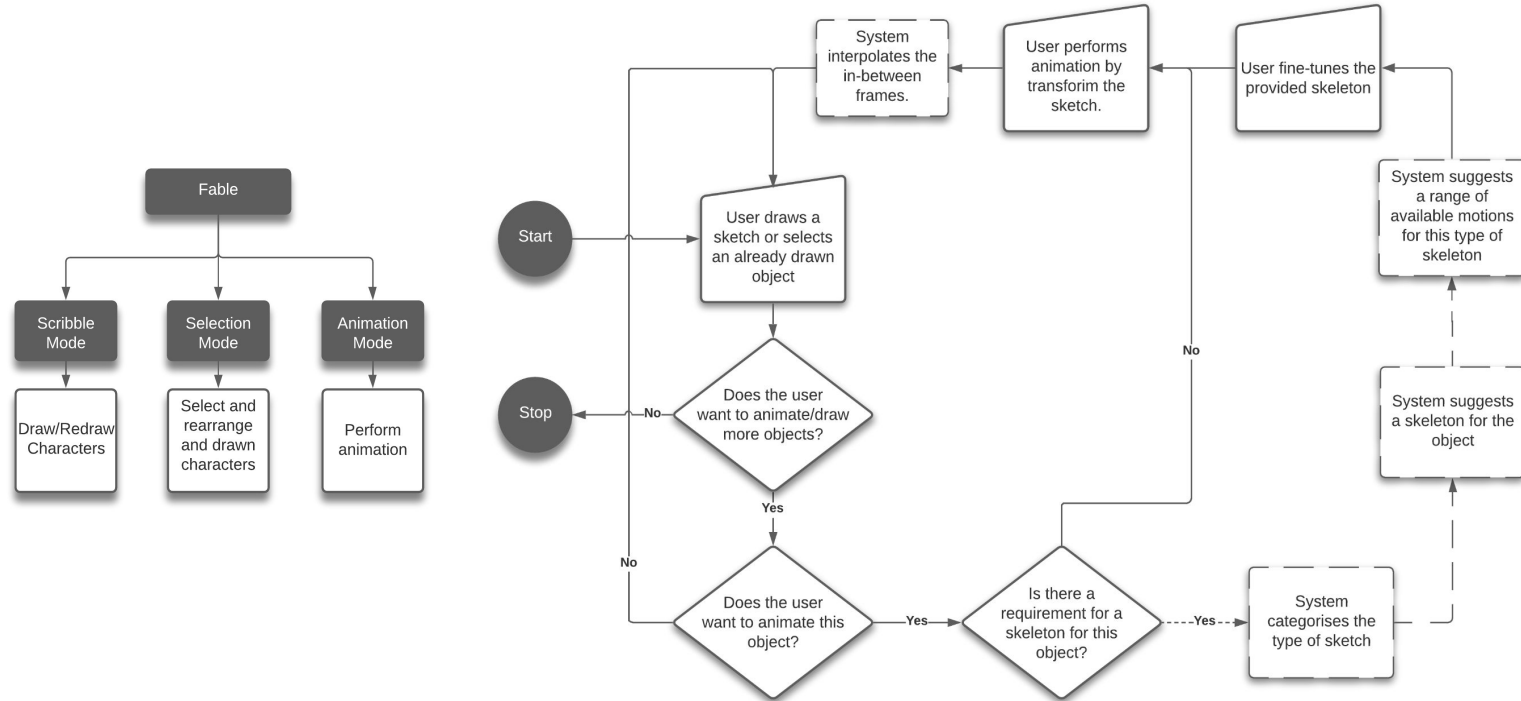
Translation

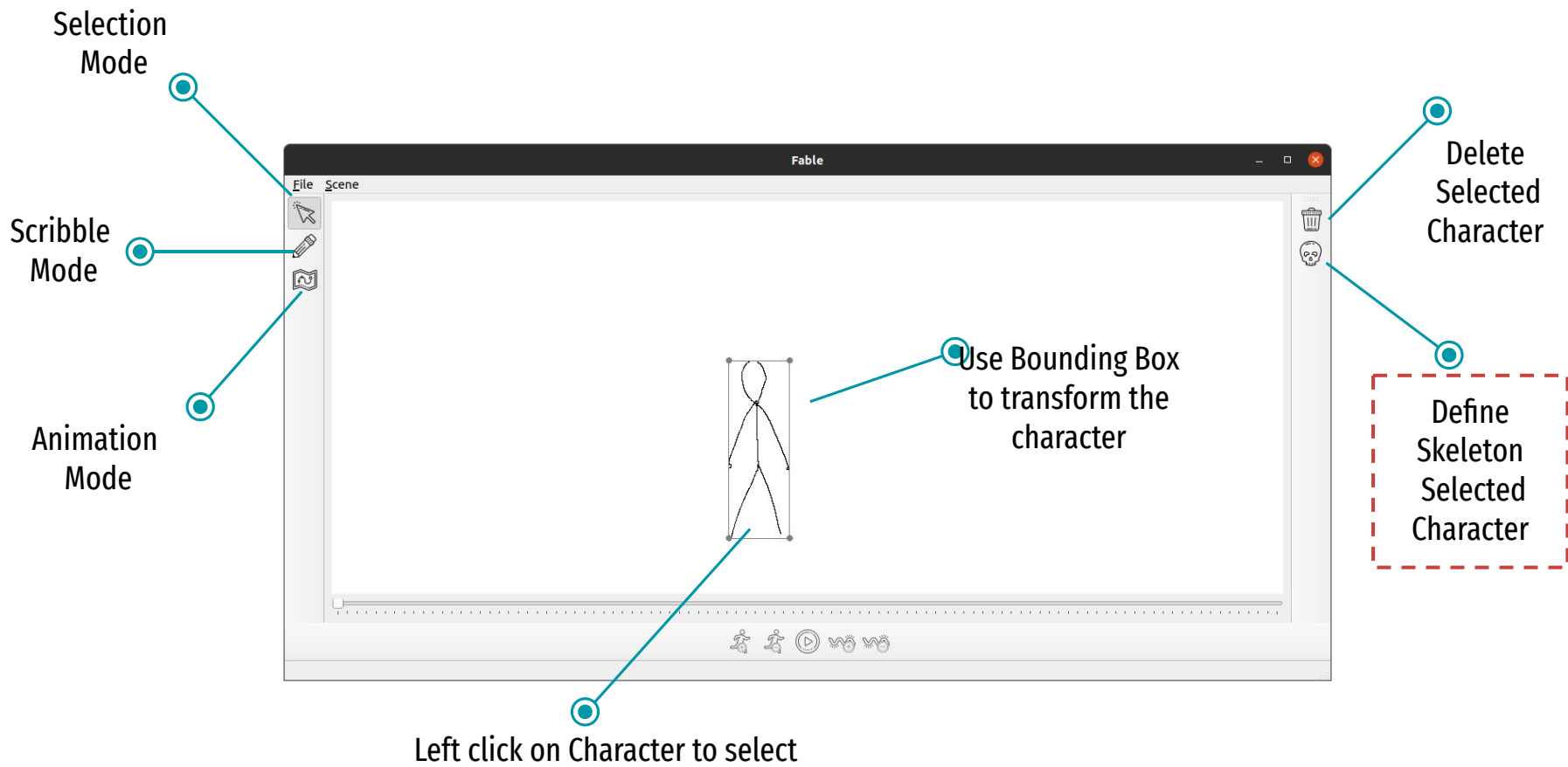


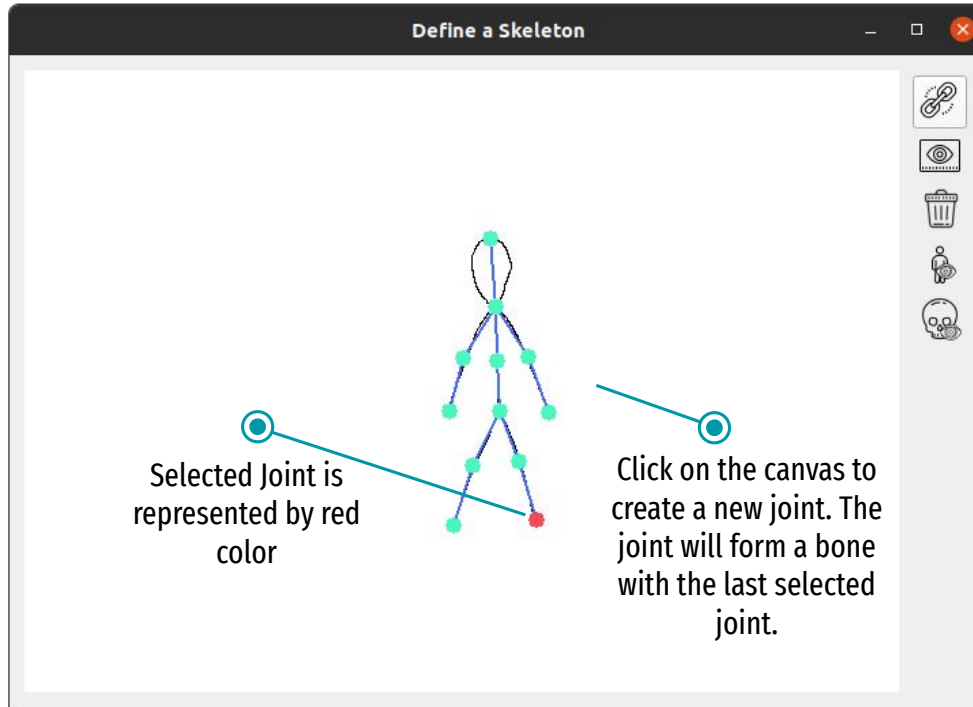
$$\Delta v = \text{lerp}(u1, v1, \Delta)$$

Live Demonstration (Skeletal Animation)

Animation Workflow







Selected Joint is
represented by red
color

Click on the canvas to
create a new joint. The
joint will form a bone
with the last selected
joint.

Bind the skeleton to character

Preview Rigging and Skinning

Delete Selected Joint

Hide/Show Skeleton

Hide/Show Cage

Selection
Mode

Scribble
Mode

Animation
Mode

Define a
new Action
Unit

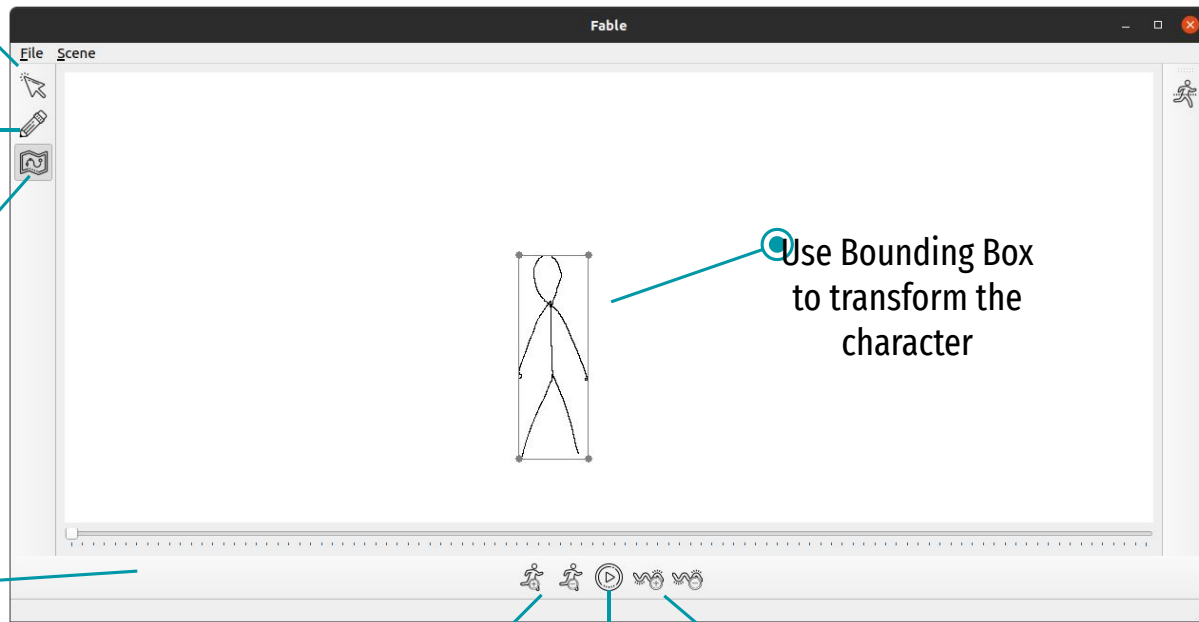
Use Bounding Box
to transform the
character

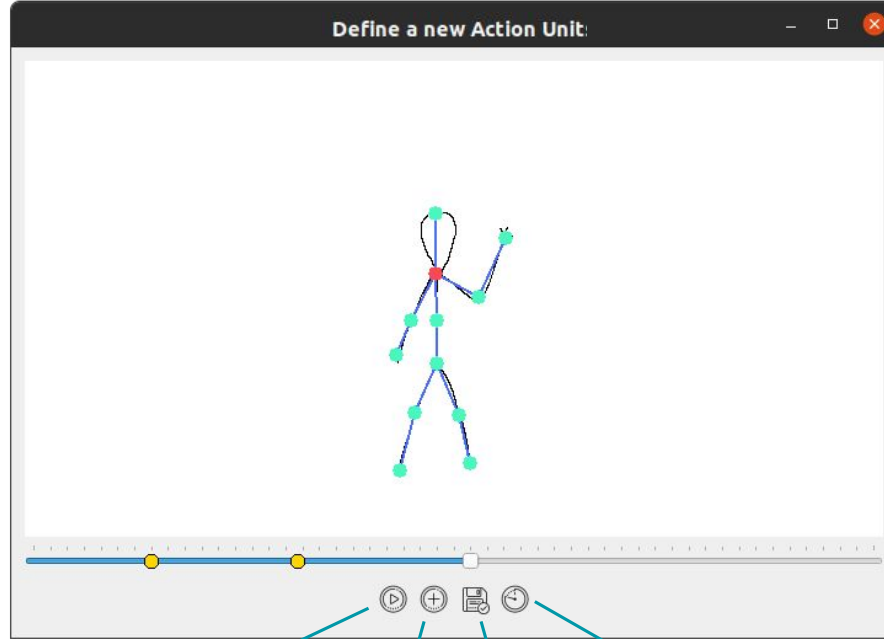
Timeline toolbar
gets activated

Add/Remove
Action Units

Play/Pause the
animation

Add/Remove
Motion Paths



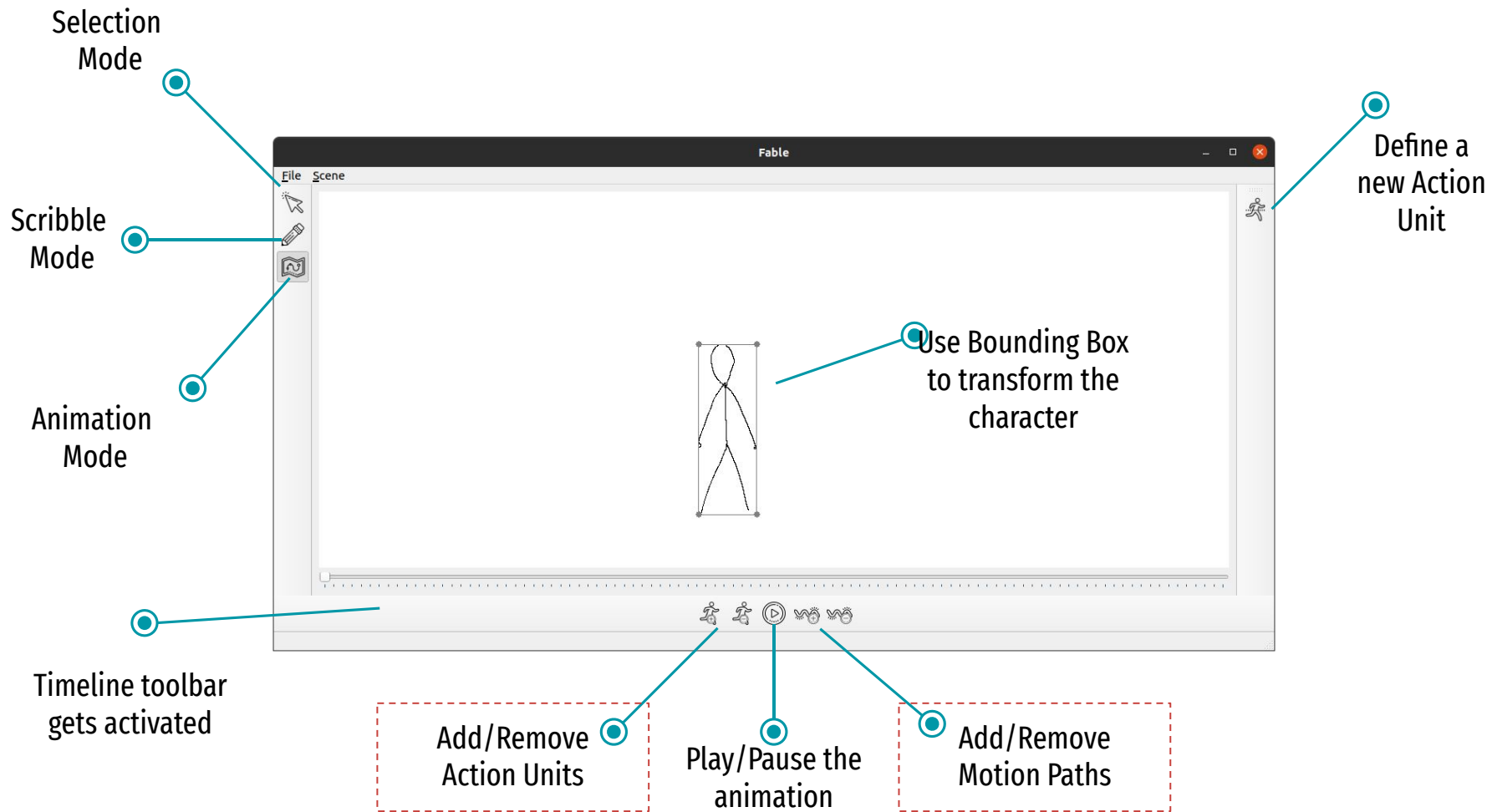


Play/Pause the
animation

Add a
Keyframe

Save the
Action Unit

Change the Duration
of Action Unit



Contributions & Acknowledgements

Contributions & Acknowledgements

Aditya's Contributions

- Linear Blend Skinning
 - Offset curve to calculate the boundary of a Sketch
 - Automatic Weight computation (from [Maurice](#))
- Skeletal Animation
 - Backend for timer management and keyframes
 - Interpolator implementation.

Shrikant's Contributions

- Manual Skeletonisation
 - Formalising the workflow
 - Rendering the Skeleton
 - Editing/Deleting Joints
 - Creating a GUI for facilitating rigging and skinning
- FABRIK- Inverse Kinematics
 - Writing a Custom Skeleton representation suitable for FABLE
 - Writing a custom implementation of Fabrik
- Animation Workflow
 - Investigating alternate workflows for animation.
 - Investigating alternate ways for implementing Action Units
 - Creating a GUI for facilitating animation.

Thank You
